Fractions, Decimals and Percentages — Paper C

HARD · NON-CALCULATOR · 50 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give fractions in their simplest form unless the question says otherwise.
- The mark for each question is shown in square brackets on the right.
- Two questions on this paper have no obvious first step. If one stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 — DECIMALS AND REVERSE PERCENTAGES

1. For each fraction, state whether its decimal terminates or recurs, and write the decimal.

(a) 3/16 [1]

(b) 5/12 [1]

(c) 7/25 [1]

You can tell before dividing. Look at the prime factors of the denominator.

2. After a 20% increase, the price of a book is 96 rupees. Find the price before the increase. [3]

3. A price is reduced by 25% to 180 rupees. Find the original price. [3]

SECTION 2 — SUCCESSIVE CHANGE

4. A sum of money is increased by 10%, and the result is then decreased by 10%.

(a) Show, using a starting value of your choice, that the final amount is not the same as the original. [2]

(b) State the overall percentage change. [1]

5. Work out 2/3 of 3/5 of 450. [3]

6. A jacket has a marked price of 2400 rupees. Two shops sell it.
Shop A: a straight 30% off.
Shop B: 1/4 off the marked price, then a further 10% off that reduced price.

(a) Find the price at each shop. [2]

(b) State which shop is cheaper, and by how much. [2]

7. A number is 5/8 of 640. Find 15% of that number. [3]

8. A bag of rice loses 12% of its mass. It now has a mass of 4.4 kg. Find the original mass of the bag. [3]

9. Write the recurring decimal 0.4444… as a fraction in its simplest form. [2]

The recurring block is one digit long. Ninths are worth knowing by heart.

10. In a sale a coat is reduced by 35%, and then a further 10% is taken off at the till. A sign says "45% off everything". Explain why the sign is wrong, and give the correct single percentage discount. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.